
Riyad Cornu

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Education

- Toulouse INP – ENSEEIHT (N7)** Toulouse, France
Engineering degree, Systems & Networks Architecture (ASR) track 2025 – 2028 (expected)
• Core curriculum: computer architecture, networks & telecommunications, language translation / compilation, concurrent & distributed systems, formal methods (Why3, Coq).
- CPGE MPI, Lycée Paul Valéry** Paris, France
Preparatory classes — Mathematics, Physics, Computer Science 2024 – 2025
- CPGE MP2I-MPI, Lycée Hoche** Versailles, France
Preparatory classes — Mathematics, Physics, Computer Science 2022 – 2024

Experience

- Basic Fit** Vélizy, France
Front Desk Agent 06/2024 – 09/2024 and 06/2025 – 09/2025
• Managed customer relations, membership sales, and after-sales support.
• Resolved conflicts and ensured proper functioning of the club facilities.
- Auchan Drive** Vélizy, France
Multi-skilled Team Member 02/2022 – 07/2022
• Prepared customer orders and managed stock and inventory levels.

Projects

- Neural Network from Scratch** 2025
• Built a neural network from scratch in CUDA, with no ML framework, including hand-written GPU kernels for every matrix operation.
• Implemented backpropagation and the Adam optimizer; reached 96% accuracy on MNIST and 89% on EMNIST.
- Unix Mini-shell** 2026
• Unix shell in C: multi-command pipelines, I/O redirection (</>), background execution (&), and builtins (cd, exit, rep).
• Process management via fork/execvp and pipes (pipe/dup2); asynchronous synchronization through a SIGCHLD handler (sigaction) and sigsuspend, with no busy-waiting or zombie processes.
- Memory Allocator (malloc)** 2026
• Reimplemented malloc/free/calloc/realloc in C directly on top of Linux primitives (mmap, sysconf), with no dependency on libc.
• Linked-list block allocator with splitting and reuse of freed regions; validated with a unit test suite (memory alignment, multiple allocations, calloc, realloc, edge cases).
- CHIP-8 Emulator** 2026
• Wrote a CHIP-8 interpreter in C, implementing the full instruction set (opcode decoding, timers, stack, registers).
• Used SDL2 for display rendering and keyboard input; validated against the IBM logo, Corax+, and BC_test ROMs.

Skills & Interests

Technical: C, Rust, Ada, CUDA (systems & low-level); Python, MATLAB (applied/scientific computing); Why3, Coq (formal methods)

Languages: French (native), English (operational)

Interests: Strength training & powerlifting, reading